

Test Case Report -- IBMS (Integrated Business Management Suite)

Field	Detail
Project Name	Integrated Business Management Suite (IBMS) v2.0
Team	Shivansh Srivastava (Product Developer), Prahallad Padhan (Product Owner), Ranveer Rai Khare (Scrum Master)
Sprint	Sprint 1
Date	April 14, 2026
Document Type	Test Case Execution Report
Test Framework	pytest (Python 3.11)
Quality Gate	100% pass rate on critical/high priority tests

1. Executive Summary

This report documents the test execution results for IBMS Sprint 1. The test suite covers **unit tests** for core security (JWT authentication) and AI services (AI assistant), as well as a comprehensive set of **planned integration and system tests** mapped to the full API surface. All implemented tests pass successfully. Additional test scenarios are identified for Sprint 2 implementation.

Metric	Value
Total Test Cases Defined	38
Tests Implemented & Executed	5
Tests Passed	5
Tests Failed	0

Metric	Value
Tests Planned (Sprint 2+)	33
Pass Rate (Executed)	100%
Code Coverage (Estimated)	~15% (security + AI assistant modules)
Target Coverage (Sprint 4)	80%

2. Test Environment

2.1 Environment Configuration

Component	Configuration
Python Version	3.11
Test Runner	pytest
Test Path	apps/ibms_core/tests/
Mock Strategy	Frappe ORM mocked via SimpleNamespace in conftest.py
Database	Mocked (no live DB required for unit tests)
CI Integration	GitHub Actions (planned)

2.2 Test Fixtures (conftest.py)

Fixture/Mock	Purpose
ROOT / APP_PATH	Adds apps/ibms_core to sys.path for module resolution

Fixture/Mock	Purpose
fake_cache	Mock Frappe cache -- get_value(), set_value() return None
fake_frappe	Mock Frappe ORM -- get_all(), cache(), session, has_permission(), throw()
Module injection	Injects mock frappe into sys.modules when not in Frappe bench

3. Test Suite Results -- JWT Authentication

Module Under Test: `ibms_core.security.jwt_auth`

Source File: `apps/ibms_core/ibms_core/security/jwt_auth.py`

Functions Tested: `issue_token()`, `validate_token()`, `decode_token()`

TC-AUTH-001: Token Issue and Validate Round Trip

Field	Detail
Test ID	TC-AUTH-001
Test Name	test_issue_and_validate_token_round_trip
Objective	Verify JWT token issuance and validation with matching secret
Preconditions	jwt_auth module imported
Steps	1. Issue token for alice@example.com (secret: secret-1, TTL: 3600s) 2. Validate token with the same secret
Expected	validate_token() returns True
Actual	True
Status	[x] PASS
Priority	Critical

Field	Detail
Type	Unit -- Positive

TC-AUTH-002: Token Payload Contains Subject and Type

Field	Detail
Test ID	TC-AUTH-002
Test Name	test_decode_token_contains_subject_and_type
Objective	Verify decoded JWT payload includes sub and token_type
Preconditions	jwt_auth module imported
Steps	1. Issue token for bob@example.com (type: access)
	2. Decode and assert sub == bob@example.com
	3. Assert token_type == access
Expected	Payload contains correct sub and token_type claims
Actual	Both assertions pass
Status	[x] PASS
Priority	Critical
Type	Unit -- Positive

TC-AUTH-003: Token Validation Fails with Wrong Secret

Field	Detail
Test ID	TC-AUTH-003

Field	Detail
Test Name	test_validate_token_fails_with_wrong_secret
Objective	Verify token rejection when validated with incorrect secret
Preconditions	jwt_auth module imported
Steps	1. Issue token for eve@example.com (secret: good-secret) 2. Validate with bad-secret
Expected	validate_token() returns False
Actual	False
Status	[x] PASS
Priority	Critical
Type	Unit -- Negative (Security)

4. Test Suite Results -- AI Assistant Service

Module Under Test: `ibms_core.services.ai_assistant`

Source File: `apps/ibms_core/ibms_core/services/ai_assistant.py`

Functions Tested: `ask_assistant()`, `recommend_actions()`

Mock Data

DocType	Mock Data
KPI Snapshot	risk_exposure: 32.1, revenue_run_rate: 12,500,000 (2026-03-13)
AI Recommendation	REC-0001, code: WF-REDUCE-APPROVAL-HOPS, confidence: 0.78

TC-AI-001: AI Assistant Risk Query

Field	Detail
Test ID	TC-AI-001
Test Name	test_assistant_risk_query
Objective	Verify AI assistant returns risk metrics for risk-related query
Preconditions	ai_assistant.frappe monkeypatched with FakeFrappe
Steps	1. Patch ai_assistant.frappe with FakeFrappe() 2. Call ask_assistant("show risk details", "Default Company", "alice@example.com") 3. Assert risk_metrics key in result 4. Assert >= 1 risk metric record
Expected	Result contains risk_metrics with >= 1 entry
Actual	risk_metrics present, 1 entry returned
Status	[x] PASS
Priority	High
Type	Unit -- Positive

TC-AI-002: AI Recommendations Payload

Field	Detail
Test ID	TC-AI-002
Test Name	test_recommendations_payload
Objective	Verify recommend_actions() returns correct recommendation count and data

Field	Detail
Preconditions	ai_assistant.frappe monkeypatched with FakeFrappe
Steps	1. Patch ai_assistant.frappe with FakeFrappe() 2. Call recommend_actions("Default Company") 3. Assert count == 1 4. Assert recommendation code == WF-REDUCE-APPROVAL-HOPS
Expected	1 recommendation returned with code WF-REDUCE-APPROVAL-HOPS
Actual	Matches expected
Status	[x] PASS
Priority	High
Type	Unit -- Positive

5. Planned Test Cases -- Authentication & Authorization

These test cases are defined for Sprint 2 implementation.

TC-AUTH-004: Login Endpoint -- Valid Credentials

Field	Detail
Test ID	TC-AUTH-004
Objective	Verify POST /api/auth/login returns access + refresh tokens with valid credentials
Type	Integration
Priority	Critical

Field	Detail
Status	Planned

TC-AUTH-005: Login Endpoint -- Invalid Password

Field	Detail
Test ID	TC-AUTH-005
Objective	Verify POST /api/auth/login returns 401 for wrong password
Type	Integration -- Negative
Priority	Critical
Status	Planned

TC-AUTH-006: Login Rate Limiting

Field	Detail
Test ID	TC-AUTH-006
Objective	Verify rate limiter blocks after excessive failed login attempts
Type	Integration -- Security
Priority	Critical
Status	Planned

TC-AUTH-007: Refresh Token Rotation

Field	Detail
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Field	Detail
Test ID	TC-AUTH-007
Objective	Verify POST /api/auth/refresh issues new token pair and invalidates old refresh token
Type	Integration
Priority	Critical
Status	Planned

TC-AUTH-008: RBAC Permission Enforcement

Field	Detail
Test ID	TC-AUTH-008
Objective	Verify viewer role cannot access POST /api/budget/optimize (admin/manager only)
Type	Integration -- Authorization
Priority	Critical
Status	Planned

TC-AUTH-009: TOTP 2FA Setup and Verification

Field	Detail
Test ID	TC-AUTH-009
Objective	Verify TOTP setup, confirm, and login with 2FA flow end-to-end
Type	Integration
Priority	High

Field	Detail
Status	Planned

TC-AUTH-010: Token Expiry Handling

Field	Detail
Test ID	TC-AUTH-010
Objective	Verify expired JWT tokens are rejected with 401 response
Type	Unit -- Negative
Priority	Critical
Status	Planned

6. Planned Test Cases -- API Endpoints

TC-API-001: Dashboard KPI Retrieval

Field	Detail
Test ID	TC-API-001
Objective	Verify GET /api/dashboard returns cached KPI snapshot with valid auth
Type	Integration
Priority	High
Status	Planned

TC-API-002: Risk Scoring

Field	Detail
Test ID	TC-API-002
Objective	Verify POST /api/risk/score returns valid risk score for transaction payload
Type	Integration
Priority	High
Status	Planned

TC-API-003: Fraud Detection

Field	Detail
Test ID	TC-API-003
Objective	Verify POST /api/fraud/detect returns requires_review: true for amount > threshold
Type	Integration
Priority	High
Status	Planned

TC-API-004: Compliance Check

Field	Detail
Test ID	TC-API-004
Objective	Verify POST /api/compliance/check evaluates controls and returns pass/fail per control

Field	Detail
Type	Integration
Priority	High
Status	Planned

TC-API-005: Budget Optimization

Field	Detail
Test ID	TC-API-005
Objective	Verify POST /api/budget/optimize returns optimized allocations within total constraint
Type	Integration
Priority	Medium
Status	Planned

TC-API-006: Dynamic Pricing Suggestion

Field	Detail
Test ID	TC-API-006
Objective	Verify POST /api/pricing/suggest returns price within base $\pm$ 50% bounds
Type	Integration
Priority	Medium
Status	Planned

TC-API-007: AI Copilot Q&A

Field	Detail
Test ID	TC-API-007
Objective	Verify POST /api/copilot/ask returns contextual AI response with confidence score
Type	Integration
Priority	Medium
Status	Planned

TC-API-008: Sales Forecast

Field	Detail
Test ID	TC-API-008
Objective	Verify POST /api/forecast returns time-series forecast with confidence intervals
Type	Integration
Priority	Medium
Status	Planned

TC-API-009: Digital Twin Simulation

Field	Detail
Test ID	TC-API-009
Objective	Verify POST /api/twin/simulate returns simulation output for valid scenario input

Field	Detail
Type	Integration
Priority	Medium
Status	Planned

TC-API-010: Inventory Prediction

Field	Detail
Test ID	TC-API-010
Objective	Verify POST /api/inventory/predict returns reorder point above safety stock level
Type	Integration
Priority	Medium
Status	Planned

7. Planned Test Cases -- ERP CRUD Operations

TC-ERP-001: Customer CRUD Lifecycle

Field	Detail
Test ID	TC-ERP-001
Objective	Verify full Create -> Read -> Update -> Delete cycle on /api/erp/customers
Type	Integration -- CRUD

Field	Detail
Priority	High
Status	Planned

TC-ERP-002: Product CRUD with Low-Stock Alert

Field	Detail
Test ID	TC-ERP-002
Objective	Verify product CRUD and that /api/erp/products/low-stock returns products below threshold
Type	Integration
Priority	High
Status	Planned

TC-ERP-003: Order Creation and Status Transitions

Field	Detail
Test ID	TC-ERP-003
Objective	Verify order creation and valid status transitions (pending -> confirmed -> shipped -> delivered)
Type	Integration -- Workflow
Priority	High
Status	Planned

TC-ERP-004: Invoice Creation and Payment Processing

Field	Detail
Test ID	TC-ERP-004
Objective	Verify invoice creation triggers anomaly check event and status update to paid
Type	Integration -- Event-Driven
Priority	High
Status	Planned

TC-ERP-005: Inventory Movement Tracking

Field	Detail
Test ID	TC-ERP-005
Objective	Verify POST /api/erp/inventory/movement records movement and updates product stock
Type	Integration
Priority	High
Status	Planned

8. Planned Test Cases -- Infrastructure & Monitoring

TC-INFRA-001: Health Endpoint

Field	Detail
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Field	Detail
Test ID	TC-INFRA-001
Objective	Verify GET /api/health returns 200 with Redis, MongoDB, MariaDB status fields
Type	Integration -- Infrastructure
Priority	High
Status	Planned

TC-INFRA-002: WebSocket KPI Stream

Field	Detail
Test ID	TC-INFRA-002
Objective	Verify WS /ws/kpi authenticates, streams KPI updates, and handles ping/pong
Type	Integration -- WebSocket
Priority	High
Status	Planned

TC-INFRA-003: Redis Failover Graceful Degradation

Field	Detail
Test ID	TC-INFRA-003
Objective	Verify system falls back to in-memory cache when Redis is unavailable
Type	Integration -- Resilience
Priority	Medium

Field	Detail
Status	Planned

TC-INFRA-004: CORS Header Validation

Field	Detail
Test ID	TC-INFRA-004
Objective	Verify CORS headers are correctly set and reject unauthorized origins
Type	Integration -- Security
Priority	Medium
Status	Planned

TC-INFRA-005: Security Headers

Field	Detail
Test ID	TC-INFRA-005
Objective	Verify responses include X-Content-Type-Options, X-Frame-Options, X-XSS-Protection, Referrer-Policy headers
Type	Integration -- Security
Priority	Medium
Status	Planned

TC-INFRA-006: Audit Log Recording

Field	Detail
Test ID	TC-INFRA-006
Objective	Verify sensitive actions (login, role change, data delete) produce audit log entries
Type	Integration -- Compliance
Priority	High
Status	Planned

9. Test Execution Summary

9.1 Results by Module

Module	Total	Passed	Failed	Skipped	Pass Rate
JWT Authentication	3	3	0	0	100%
AI Assistant Service	2	2	0	0	100%
Total (Executed)	5	5	0	0	100%

9.2 Results by Priority

Priority	Total	Passed	Planned	Pass Rate (Executed)
Critical	10	3	7	100%
High	18	2	16	100%
Medium	10	0	10	--

Priority	Total	Passed	Planned	Pass Rate (Executed)
Total	38	5	33	100%

9.3 Results by Type

Type	Executed	Planned
Unit -- Positive	4	0
Unit -- Negative	1	1
Integration	0	19
Integration -- Security	0	4
Integration -- CRUD	0	1
Integration -- Workflow	0	1
Integration -- Event	0	1
Integration -- Resilience	0	1
Integration -- Compliance	0	1
Integration -- WebSocket	0	1
Integration -- Authorization	0	1
Integration -- Infrastructure	0	1

10. Defects Found

No defects were found during Sprint 1 test execution. All 5 executed tests passed on first run.

11. Test Coverage Gaps & Recommendations

Gap Area	Current State	Recommendation	Sprint Target
Auth endpoint integration tests	No integration tests for login/register/refresh	Implement TC-AUTH-004 through TC-AUTH-010	Sprint 2
API endpoint coverage	Services tested only via unit mocks	Add FastAPI TestClient integration tests	Sprint 2
ERP CRUD operations	0% coverage	Implement TC-ERP-001 through TC-ERP-005	Sprint 2
WebSocket testing	0% coverage	Add async WebSocket test using websockets library	Sprint 2
Database layer	Only mocked	Add integration tests with test MongoDB/MariaDB containers	Sprint 3
Security headers	Not validated	Automate header validation in CI	Sprint 2
Performance/load	No load tests	Add locust or k6 load tests for critical endpoints	Sprint 3
ML model accuracy	Not measured	Add model evaluation tests with labeled datasets	Sprint 3

12. Approval

Role	Name	Date
Product Developer	Shivansh Srivastava	April 14, 2026
Product Owner	Prahallad Padhan	April 14, 2026

Role	Name	Date
Scrum Master	Ranveer Rai Khare	April 14, 2026